



ADVANCING SMART TRAFFIC INFRASTRUCTURE WITH AI-BASED VEHICLE DETECTION ACROSS COMPUTING CONFIGURATIONS

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ABSTRACT

Resilient, quality, sustainable and reliable infrastructure are dominant to enhance urban transportation as aligned with Sustainable Development Goals (SDG) 9 and 11 by promoting sustainable industrialization. Existing research merely focuses on evaluating the performance of detection models across different techniques and technologies. This study aims to examine the variations in deep learning model design and computational hardware in influencing vehicle detection performance. Several segmentation and detection models were trained and evaluated under different preprocessing techniques and Graphics Processing Units (GPU) environments. The results show that blurring irrelevant regions remarkably enhanced the detection performance while heavy augmentation increased the training time despite maintaining accuracy. Hardware advancements greatly reduced training time with only minor improvement in detection. The findings emphasize the importance of equipping computing configurations to build a more scalable and resilient traffic surveillance system that can support smart city initiatives through accurate and faster vehicle detection.

Keywords: Vehicle Detection; Deep Learning; Smart Traffic Infrastructure; Sustainable Urban Mobility

INTRODUCTION

Building a smart and resilient traffic infrastructure for a better smart city framework always requires innovative breakthroughs from multiple authorities to further improve the urban transportation system. In the alignment with SDG 9 and 11 which draws attention to promoting sustainable industrialization within the community, vehicle detection models that are powered by sophisticated technologies such as Internet of Things (IoT) come to be important in achieving these objectives across the countries [1], [2]. Having effective detection models is instrumental in building more responsive infrastructure which could power up infrastructure development. The integration of machine learning (ML) in vehicle detection could bolster the creation of smart city frameworks by making full use of the resource and improving energy efficiency [3], [4]. The advent of technologies such as Light Detection and Ranging (LiDAR) sensors technology also come hand in hand in boosting long-range detection performance due to its ability to provide 360-degree surveillance and generate 3D maps of subject areas which make them suitable for detecting vehicles under various traffic conditions [5]. Moreover, the use of infrared sensors combined with deep learning algorithms has shown great improvement in achieving higher detection accuracy [6]. Building powerful vehicle detection models in the automobile industry could have a great impact on several aspects, including development of Electric Vehicles (EV) and intelligent car driving [4], parking lot management, license plate recognition [7] and even traffic accidents scene reconstruction [8].

Upon literature research, the authors identified that there are indeed numerous studies that emphasize the techniques and technologies aiming to achieve more accurate and faster vehicle detection models by focusing on various algorithms, performance metrics, interdisciplinary technological collaborations and system deployment. However, there exists a lack of research on how both software and hardware settings could make an impact to the overall model efficiency and their suitability to contribute to a more resilient land transport infrastructure. In light of this, this study aims to investigate the impact of model variations and computational



hardware on the deep learning-aided vehicle detection for more scalable traffic surveillance focusing on You Only Look Once (YOLO) algorithm. The detailed analysis will focus on evaluating the capability of the trained models in assisting intelligent transportation systems by boosting vehicle detection accuracy.

METHODOLOGY

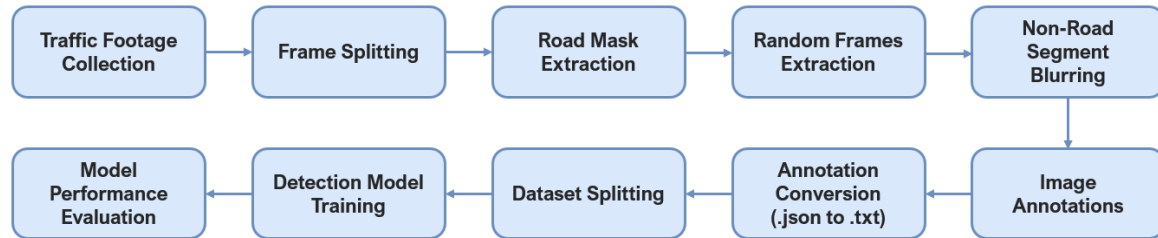


Figure 3 – Vehicle detection model workflow diagram.

DATA COLLECTION AND PREPROCESSING

The workflow diagram examining the performance of various vehicle detection models is presented in Figure 1. The development of vehicle detection model began with data collection, where traffic footages were captured from the Shah Alam Expressway (KESAS Expressway), a 37.5-mile highway in Selangor, Malaysia. 78 traffic footages were taken with Full High Resolution (FHD) from 23rd December 2024 to 13th January 2025 lasting for 21 days, and these footages covered various conditions including time of day (daytime to nighttime recordings), weather conditions (sunny, foggy, raining, clear and cloudy conditions), various levels of congestion and days of week (weekdays, weekends and public holidays) for the betterment of training and detecting the images.

With the objective of protecting privacy for both vehicles and pedestrians, all video and image processing were carried out offline using customized Python scripts for automating the processing procedures to completely eliminate any potential risks upon online processing. For frame extraction and annotation, open-source command-line tool *FFmpeg* was utilized for splitting frames based on the specified frame per second (fps), while graphical annotation tool *LabelMe* version 5.5.0 assisted the road segments extraction. Non-road areas were also blurred accordingly using Gaussian Blur to reduce background noise with its ability to provide gradual and smooth blurring and better edge preservation [9]. As a result, there was a total of 2208 images for annotations, which were then divided into 1545 training images, 444 validation images and 219 test images according to the ratio train:valid:test of 70:20:10. Polygon annotations have been a preferred choice to annotate object boundaries and segmentation masks as it allows the dataset to be used for both object detection and instance segmentation model training.

MODEL TRAINING AND EVALUATION

There were 6 different vehicle detection models trained for assessing the comparison in terms of both software factors – model types, dataset augmentation strategies and non-road segment blurring, as well as hardware factors – GPU performance. YOLO algorithm was selected with its excellent performance in both object detection and instance segmentation, and Google Colab Pro platform with Python 3.11.11 has been utilized for model training due to its efficiency in training large datasets. All 6 detection models were trained under the same configuration: 100 epochs with a batch size of 32. To evaluate the impact of data augmentation on model performance, horizontal flip, vertical flip and image gray-scaling were taken into consideration. These augmentation techniques were randomly applied to each image within the dataset. This resulted in 4753 images consisting of 3324 training images, 957 validation images and 472 test images. The performance comparison was evaluated specifically for mean Average Precision (mAP), precision, recall, F1-score and training duration to examine the ability of the model towards contributing to a smarter and responsive traffic



infrastructure. The details of each model trained are expressed as in Table 1. The bolded cells in the table indicate the variations where the model differs from the baseline. The baseline model was constructed with straightforward configuration: instance segmentation with blurred non-road regions, no dataset augmentation and T4 GPU. This setting serves as a minimal reference point for examining the subsequent variation in model architecture, preprocessing and hardware to ensure the performance is measurable and impactful.

Table 6 – Model training with software and hardware variations.

Model	Model Type	Non-Road Regions	Augmentation	Hardware	Remarks
1	Instance Segmentation	Blurred	No	T4 GPU	Baseline model
2	Object Detection	Blurred	No	T4 GPU	
3	Instance Segmentation	Unblurred	No	T4 GPU	Software variations
4	Instance Segmentation	Blurred	Yes	T4 GPU	
5	Instance Segmentation	Blurred	No	L4 GPU	Hardware variations
6	Instance Segmentation	Blurred	No	A100 GPU	

RESULTS AND DISCUSSIONS

The performance statistics of trained detection models were documented with the results categorized according to software and hardware variations as presented in Table 2.

Table 7 – Model training results visualization.

Model	mAP50	mAP50-95	Precision	Recall	Accuracy	F1 Score	Duration (Hours)
1	0.9620	0.7720	0.9123	0.9600	0.8789	0.9355	2.836
2	0.9620	0.7830	0.9051	0.9584	0.8709	0.9310	2.329
3	0.4020	0.2980	0.5210	0.6197	0.3948	0.5661	3.077
4	0.9600	0.7750	0.9119	0.9559	0.8746	0.9331	5.987
5	0.9610	0.7790	0.9157	0.9548	0.8777	0.9348	0.916
6	0.9600	0.7770	0.9138	0.9557	0.8777	0.9343	0.549

It was observed that the baseline model consistently outperforms other testing models across the performance metrics. A notable observation is that both segmentation and detection models achieved similarly high mAP50 scores when non-road regions were blurred, which means that reducing visual noise within the dataset could enhance the detection performance regardless of model type. While the detection model (Model 2) achieved slightly higher mAP50-95 than the segmentation model (Model 1), the overall differences in precision, recall, accuracy and F1-score were minimal. In this context, segmentation models still remain a strong option, especially for tasks that require concise boundary detection such as lane markings or multi-class traffic monitoring. Apart from that, the applied dataset augmentation strategy appeared to have a subtle impact in a way it maintained high performance, but it incurred almost double the training duration (Model 4). This observation makes us question whether having heavy data augmentation operation is worth the extra time it requires when it comes to implementing in real-world settings where faster model training time is preferred.

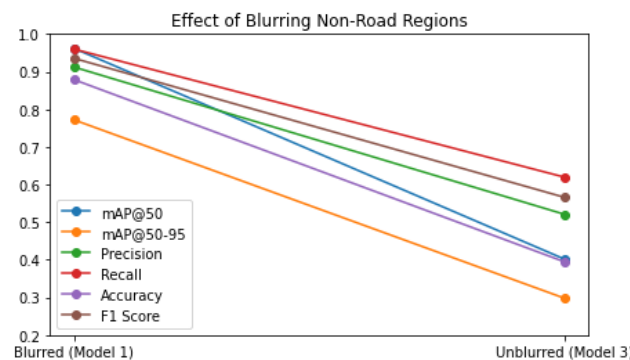


Figure 4 – Effects of blurred and unblurred non-road regions



As depicted in Figure 2, non-road region handling serves as one of the important metrics in evaluating the effectiveness towards smart traffic infrastructures. Model trained with non-relevant areas, focusing on only vehicle-labeled regions, achieved drastically higher performance (Model 1) across all examined metrics compared to those trained with unblurred images (Model 3). Blurring irrelevant areas could help improve accuracy while also reduce computational costs during training.

In terms of hardware impact, moving from a T4 GPU to an L4 (Model 5) or A100 (Model 6) showed only slight increment in terms of mAP50-95, but it resulted in drastic reduction in training duration especially for A100 GPU. While the model accuracy plateaus to a certain extent, having advanced and decent hardware could speed up the training which is an important factor when frequent model updates are needed in urban settings. Overall, the findings show that deep learning-based vehicle detection could effectively contribute to the advancement of smart traffic infrastructure by having more accurate and faster vehicle counting, traffic flow prediction or even congestion management.

CONCLUSIONS

This study explored the impact on both model variations and computational hardware on AI-based vehicle detection for smarter traffic infrastructure. In balancing the trade-off between model performance and training duration, this challenge could be overcome by combining both software and hardware strategies to develop a highly effective model suitable to be deployed in real-world scenarios. For instance, combining dataset augmentation methods with an A100 GPU can offset the downsides of long training time while also boosting the detection performance. Paving way to contribute to an intelligent transportation system, future work could focus on deploying the model into broader traffic applications such as a traffic prediction system or a traffic surveillance framework to analyze traffic condition within a specific region. The model performance could also be further investigated in future research with more decent hardware configurations such as Tensor Processing Units (TPU). All in all, this research lays the foundation of having accurate and faster vehicle detection framework, which can lead to informed decisions for researchers and urbanists about the future infrastructure planning to ensure the roads and transportation systems are designed to meet future needs.

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